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On the Singularities in the Dynamics of a Spherical Top with Nonholonomic Constraints

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Abstract—The problem of the motion of a dynamically symmetric ball with a shifted center of gravity on a horizontal absolutely rough plane is considered, taking into account the unilateral nature of the contact. It is shown that for some values of the parameters and initial conditions, singularities of two types may appear: ambiguity in determining the subsequent motion (separation of the body from the support or continuation of rolling) or the impossibility of determining it within the framework of the model used (the finite time paradox). For a formal description of these situations, the generalized complementarity problem is used. The Littlewood problem of the rolling of a hoop with a point mass is studied in detail. Some cases of paradoxes when imposing differential constraints of various types are also discussed, including the generalization of the Chaplygin sleigh and the “rubber” ball.

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1. INTRODUCTION

The dynamics of a rigid body on a horizontal plane is one of the most famous classical problems in mechanics. The key point in describing the motion is the choice of a contact interaction model. The generally accepted requirements for a mathematical model — adequacy, correctness, accuracy, simplicity, robustness, efficiency, etc. — are often contradictory. In Multibody Dynamics, a rigid body model is used, and the Amontons – Coulomb law is used to describe tangential stresses (in the limiting cases of zero and infinite friction coefficients — an absolutely smooth and absolutely rough support). These simple models do not, generally speaking, have the property of correctness: even in the absence of friction, there are examples of nonuniqueness of the solution to the equations of motion [1, 2]. Examples of nonuniqueness in systems with Coulomb static friction were first discovered by Jellett [3]. In [4, 5], an example of the coexistence of three solutions at once is given: separation of the body from the support, continued rolling, and the beginning of sliding. Much more famous were Painlevé’s later studies of the sliding friction case: here both nonuniqueness and absence of solutions are possible under certain initial conditions [6–8]. It is important to note that such examples were constructed for both types of connection between the body and the support: unilateral and bilateral contact. Subsequently, situations of this kind became known as Painlevé paradoxes. They are caused by the incompatibility of the rigid body model with the friction law. Correctness can be achieved by taking into account small deformations or by correcting the

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Amontons – Coulomb law, but in this case it is necessary to supplement the problem statement with some physical concepts, the choice of which is ambiguous [9, 10]. It was proposed to describe friction by Coulomb's law taking into account the length of the contact region (Contensou – Zhuravlev model [11–13]), but in this model a singularity arises in the case of simultaneous vanishing of the angular velocity of rotation and the sliding velocity, which is expressed in an abrupt change in the integral friction force with small increments of these velocities. Rolling systems can intuitively be considered a limiting case of a large coefficient of friction, excluding sliding at the point of contact. Here, the assumption of the possibility of the body being torn away from the support is essential. If such a possibility is excluded at the stage of problem formulation, then the contact conditions can be described using a pair of differential constraints. In this case, the highly developed apparatus of nonholonomic mechanics is used for the study (see [14–16] and references therein).

A more natural idea of a unilateral contact between a body and a plane leads to a conflict with the assumption of no slip. Apparently, Littlewood [17] was the first to draw attention to this, showing for a particular case that a nonuniform disk cannot roll on a support without slipping at an arbitrarily high angular velocity. At some point, a contradictory situation arises when neither separation from the support nor continuation of rolling is possible. This behavior is sometimes called the "finite time paradox". As experiments have shown [18–20], in reality the disk will first begin to slide and then jump. A number of authors have suggested the existence of a "gliding" regime to resolve the paradox: there is no separation, the normal reaction disappears, and the friction force remains finite [21]. Obviously, such a solution is inconsistent with the original assumption of no slip and has not been confirmed experimentally.

Along with the Littlewood paradox, a "hidden jump" type singularity is possible in systems of this type [22, 23]: the body can either continue to roll along the support or break away from it under the same conditions. Ref. [24] provides an example of regular precession, for which, along with continuous motion, the body can also break away from the support at any moment in time. Ref. [22] examines periodic motions of a nonuniform disk on a curvilinear rough support and identifies segments of trajectories in which a "hidden jump" is possible.

From a formal point of view, the absence of sliding is expressed by imposing nonholonomic constraints on the system. In the classical case, there are two such constraints, i. e., their number is equal to the dimension of the tangent space. Systems with three constraints are also known — the so-called "rubber" ball, which cannot slide or even spin (see [28], where the history of the problem is also given). The papers [25–27] investigate a problem with a single constraint describing the prohibition of sliding in a certain direction.

From a practical point of view, the study of the uniqueness of the continuous motion is important, in particular when designing a robot-ball, since singularities remain unnoticed in computer modeling. Such an analysis was carried out, in particular, in [23], where a general approach to such problems was outlined, an example of the singularity of the precession of a top was constructed, and the case of a robot with a variable radius of gyration was considered. C. Liu et al. [29] performed a theoretical and experimental study of bouncing motions in a robotic system. Various methods for designing and controlling spherical robots are considered in [30].

In this paper, the dynamics of a rigid body rolling on a plane are discussed using the example of a spherical top, i. e., a dynamically symmetric ball with a shifted center of gravity. The cases of one, two and three nonholonomic constraints are considered. In all of the above cases, the constraint between the body and the support is considered unilateral. The conditions for the occurrence of singularities of the "finite time paradox" and "hidden jump" types are obtained.

2. SYSTEM FORMULATION

2.1. Equations of Motion

Consider a dynamically symmetric rigid body with a spherical surface and center of mass on the axis of symmetry, moving on a horizontal plane. We introduce an inertial coordinate system $OXYZ$ with origin in the plane and the axis OZ directed upwards. Let i_1, i_2, i_3 denote the unit vectors of these axes. Let G be the center of mass. We will attach to the body the system Ge_1, e_2, e_3, with Ge_3 directed along the symmetry axis.

As generalized coordinates $q \in \mathbb{R}^6$ we choose the Cartesian coordinates (x, y, z) of the point G and the Euler angles θ, ψ, φ. Let us write the equations of motion with Lagrange multipliers [31]:

$$\begin{gathered}\frac{d}{dt}\left(\frac{\partial T}{\partial \dot q_k}\right) - \frac{\partial T}{\partial q_k} = Q_k + R_k, \quad R_k = \sum_{i=1}^{l} \lambda_i \gamma_{ik}, \quad k = 1, \ldots, 6, \\ T = \frac{1}{2} m \left(\dot q_1^2 + \dot q_2^2 + \dot q_3^2\right) + \frac{1}{2}\left(J_1\left(p^2 + q^2\right) + J_3 r^2\right)\end{gathered} \tag{2.1}$$

where T is the kinetic energy of the body, m is its mass, J_1 and J_3 are the equatorial and axial moments of inertia, R_k denote the reactions of nonholonomic constraints, λ_i are the Lagrange multipliers, and γ_{ik} are the coefficients in their expressions:

$$\sum_{j=1}^{6} \gamma_{ij} \dot q_j = 0, \quad i = 1, \ldots, l, \tag{2.2}$$

l being the number of such constraints. Below we will consider the cases $l = 1, 2, 3$. The system (2.1)–(2.2) is closed with respect to the unknown functions $q_k(t)$, $\lambda_i(t)$, and its solutions describe the dynamics of the body in contact with the support. The projections of the angular velocity onto the moving axes are calculated using the formulas

$$p = \dot\psi \sin\theta \sin\varphi + \dot\theta \cos\varphi, \quad q = \dot\psi \sin\theta \cos\varphi - \dot\theta \sin\varphi, \quad r = \dot\psi \cos\theta + \dot\varphi. \tag{2.3}$$

The unilateral nature of the contact is expressed by the inequality

$$\Lambda(q) = z - f(\theta) \geqslant 0, \quad f(\theta) = \rho + a\cos\theta, \tag{2.4}$$

where ρ is the radius of the sphere and $a = |OG|$ is its eccentricity. The generalized forces Q_k depend on the external forces applied to the body as well as on the normal reaction of the support N. In particular, if there are no other active forces, then $Q_3 = N - mg$, $Q_4 = aN \sin\theta$, and all other generalized forces are equal to zero. Reactions R_k depend on the imposed nonholonomic constraints according to (2.1).

2.2. Dynamics Singularities

The presence of unilateral constraint (2.4) determines the existence of different types of body motion: if $\Lambda(q) \equiv 0$ over a certain time interval, then the body rolls or slides on the support, but if $\Lambda\big(q(t_0)\big) = 0$, $\dot\Lambda\big(q(t_0)\big) = 0$, $\ddot\Lambda\big(q(t_0)\big) > 0$, then it becomes airborne. From a formal point of view, to describe the second type of motion, one can use Eqs. (2.1), in which one must put $R = 0$. In addition, for physical reasons (there is no adhesion), the normal reaction is nonnegative, i. e., $N = (R, i_3) \geqslant 0$. The pair $\ddot\Lambda$ and N is dual in the linear complementarity problem (LCP) [32]:

$$\begin{gathered}\ddot\Lambda = MN + b, \\ \ddot\Lambda \geqslant 0, \quad N \geqslant 0, \quad \ddot\Lambda N = 0.\end{gathered} \tag{2.5}$$

As is well known, the criterion for the existence of a unique solution in problem (2.5) is that the following inequality is satisfied:

$$M > 0. \tag{2.6}$$

In this case, the nature of the motion in the interval $t > t_0$ is determined by the value of b: if $b \leqslant 0$, then the contact persists, and if $b > 0$, then it is interrupted. In the case where inequality (2.6) has the opposite value, paradoxical situations of ambiguity (if $b > 0$) or nonexistence (if $b < 0$) of the solution to problem (2.5) arise.

As shown in [23], in rolling systems the LCP does not cover all physically possible situations, since in the case of separation not only the normal but also the tangential component of the reaction disappears. Here the generalized linear complementarity problem (GLCP) should be used in the form

$$\ddot{\Lambda} = \begin{cases} MN + b, & \text{if } \Lambda = 0, \\ b', & \text{if } \Lambda > 0, \end{cases} \quad \ddot{\Lambda} \geqslant 0, \quad N \geqslant 0, \quad \ddot{\Lambda} N = 0. \tag{2.7}$$

The following statement is true.

Theorem (see [23]). *GLCP* (2.7) *has a unique solution iff $Mbb' > 0$. In the case of $Mb > 0$, $b' < 0$ there is no solution. If $Mb < 0$, $b' > 0$, then there are two solutions. A full list of possible cases is given in Table* 1: *here the symbol $\nearrow$ denotes the loss of contact of the body with the support, C indicates the continued rolling, and $\varnothing$ means that no solution exists.*

Table 1. Variants of solution to GLCP

b'	M, b			
	$+\,+$	$+\,-$	$-\,+$	$-\,-$
$+$	$\nearrow$	$\nearrow$ C	$\nearrow$ C	$\nearrow$
$-$	$\varnothing$	C	C	$\varnothing$

The proof of this statement is carried out by a straightforward enumeration of various possible combinations of signs.

Note that in computer studies of the dynamics of a rigid body in contact with a support, a simplified approach is often used, considering the inequality $N > 0$ sufficient for the contact to persist, and the change of the sign of this inequality as a condition for the loss of contact. However, these considerations are valid only under the condition $Mbb' > 0$, which has not been verified by anyone. To verify it, we calculate $\ddot{\Lambda}$ in formula (2.4):

$$\begin{gathered} \dot{\Lambda}(q) = \dot{z} - f'(\theta)\dot{\theta}, \quad \ddot{\Lambda}(q) = \ddot{z} - f'(\theta)\ddot{\theta} - f''(\theta)\dot{\theta}^2, \\ f' = -a\sin\theta, \quad f'' = -a\cos\theta. \end{gathered} \tag{2.8}$$

In this case, the second derivatives $\ddot{z}$ and $\ddot{\theta}$ in formula (2.7) are calculated from the Lagrange equations (2.1):

$$m\ddot{z} = N - mg + R_3, \quad J_1\ddot{\theta} - J_1\dot{\psi}^2\sin\theta\cos\theta + J_3(\dot{\psi}\cos\theta + \dot{\varphi})\sin\theta = aN\sin\theta + R_4.$$

Substituting these values into LCP (2.5), we obtain

$$\ddot{\Lambda} = \frac{N}{m} - g + \frac{a^2}{J_1}N\sin^2\theta + a\dot{\theta}^2\cos\theta + a\sin\theta\left(\dot{\psi}^2\sin\theta\cos\theta - \frac{J_3}{J_1}(\dot{\psi}\cos\theta + \dot{\varphi})\dot{\psi}\sin\theta\right) + \frac{R_3}{m} + \frac{R_4}{J_1}.$$

Thus,

$$\begin{gathered} M = \frac{1}{m} + \frac{a^2}{J_1}\sin^2\theta + \frac{R_3^N}{m} + \frac{R_4^N}{J_1}, \\ b = a\sin^2\theta\left(\dot{\psi}^2\cos\theta - \frac{J_3}{J_1}(\dot{\psi}\cos\theta + \dot{\varphi})\dot{\psi}\right) + a\dot{\theta}^2\cos\theta - g + \frac{R_3^0}{m} + \frac{R_4^0}{J_1} \end{gathered} \tag{2.9}$$

where $R_i = R_i^N N + R_i^0$. During a takeoff, $N = 0$, $R_3 = R_4 = 0$. Hence,

$$\begin{gathered} b' = a\sin^2\theta\left(\dot{\psi}^2\cos\theta - \frac{J_3}{J_1}k\dot{\psi}\right) + a\dot{\theta}^2\cos\theta - g, \\ \dot{\psi} = \frac{k - J_3 r\cos\theta}{J_1\sin^2\theta}. \end{gathered} \tag{2.10}$$

In this formula, the change in the angular velocities is described by Euler's dynamical equations

$$\begin{cases} J_1\dot{p} + (J_3 - J_1)qr = 0, \\ J_1\dot{q} + (J_1 - J_3)pr = 0, \\ J_3\dot{r} = 0 \end{cases}$$

which have two first integrals (see formulas (3.11) below), which allows us to reduce this system to a single equation with respect to θ. A body that has lost contact with the supporting plane performs a regular precession.

Conclusion: the correctness of GLCP depends on the reactions of nonholonomic constraints imposed on the system. In particular, if there are no such constraints (the supporting plane is absolutely smooth), then $M > 0$, $b' = b$, and the behavior of the top is uniquely determined by the sign of the value of b: if it is negative, then the contact of the body with the support persists, and in the case of $b > 0$ it is destroyed.

2.3. *Littlewood's Example [17]*

Consider a nonhomogeneous symmetrical spherical top of unit mass and radius ρ rolling without spinning in the fixed vertical plane containing the axes of symmetry on a perfectly rough support (Fig. 1). Let O denote the center of the sphere, let G and C be its center of mass and the point of contact, and $a = |OG|$ be the eccentricity.

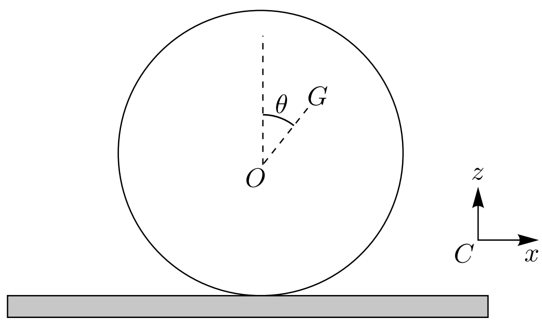

Fig. 1. Littlewood's example.

When in contact with the support, the system is subject to the unilateral constraint $z \geqslant \rho + a\cos\theta$, as well as the differential (integrable) constraint $\dot{x} - (\rho + a\cos\theta)\dot{\theta} = 0$, expressing the absence of slip. With these constraints, the kinetic and potential energy have the form

$$T = \frac{1}{2}\left(\rho^2 + a^2 + J + 2a\rho\cos\theta\right)\dot{\theta}^2, \quad \Pi = g(\rho + a\cos\theta).$$

Let us formulate the Lagrange equation for this system:

$$\left(\rho^2 + a^2 + J + 2a\rho\cos\theta\right)\ddot{\theta} - a\rho\dot{\theta}^2\sin\theta - ag\sin\theta = 0.$$

Next, we perform calculations using formula (2.8):

$$\Lambda = z - (\rho + a\cos\theta), \quad \ddot{\Lambda} = N - g + a\ddot{\theta}\sin\theta + a\dot{\theta}^2\cos\theta,$$
$$\ddot{\theta} = \frac{\rho\dot{\theta}^2 + g}{\rho^2 + a^2 + J + 2a\rho\cos\theta} a\sin\theta. \tag{2.11}$$

Therefore,

$$M = 1, \quad b = \dot{\theta}^2\left(a\cos\theta + \frac{a^2\rho\sin^2\theta}{(\rho^2 + a^2 + J + 2a\rho\cos\theta)}\right) + g\left(\frac{a^2\sin^2\theta}{\rho^2 + a^2 + J + 2a\rho\cos\theta} - 1\right). \tag{2.12}$$

Following [17], we assume that $a = \rho$, $J = 0$, while

$$M = 1, \quad b = \dot{\theta}^2 \left(\rho \cos\theta + \frac{\rho \sin^2\theta}{2(1+\cos\theta)} \right) + g\left(\frac{\sin^2\theta}{2(1+\cos\theta)} - 1 \right),$$
$$T = \rho^2(1+\cos\theta)\dot{\theta}^2, \quad \Pi = g\rho(1+\cos\theta).$$

If at the initial time $\theta = \theta_0 = 0$, $\dot{\theta} = \dot{\theta}_0 \ll 1$, then the total energy of the top is equal to

$$E = T + \Pi = 2g\rho$$

(the value $\dot{\theta}_0$ is neglected). Since $E = \text{const}$,

$$\rho^2(1+\cos\theta)\dot{\theta}^2 + g\rho(1+\cos\theta) = 2g\rho \quad \Rightarrow \quad \dot{\theta}^2 = \frac{g(1-\cos\theta)}{\rho(1+\cos\theta)}.$$

Therefore,

$$b = g\frac{1-\cos\theta}{1+\cos\theta}\left(\cos\theta + \frac{1-\cos\theta}{2}\right) + g\left(\frac{1-\cos\theta}{2} - 1\right) = -g\cos\theta.$$

In the airborne phase $\ddot{z} = -g$, $\dot{\theta} = \omega = \text{const}$, whence

$$b' = \rho\omega^2\cos\theta - g.$$

During a hypothetical "flight", the magnitude of the angular velocity is constant:

$$\omega^2 = \frac{g(1-\cos\theta_1)}{\rho(1+\cos\theta_1)}$$

where θ_1 is the value at the time of takeoff.

Thus,

$$b' = \frac{g(1-\cos\theta_1)}{1+\cos\theta_1}\cos\theta - g < 0. \tag{2.13}$$

Inequality (2.13) is satisfied in two cases: either $\cos\theta \leqslant 0$ or $\cos\theta > 0$, $\cos\theta_1 > 0$. At the instant of takeoff, $\theta = \theta_1$, and hence it is valid.

We can conclude that at first the top rolls on the support, and when the variable θ has reached the value $\frac{\pi}{2}$, the function $b(\theta)$ vanishes, and it is possible neither to continue rolling $\big($since for $\theta > \frac{\pi}{2}$ we have $b > 0\big)$ nor to take off due to (2.13).

Note 1. *This problem introduces a "finite time paradox" that is not described in the LCP framework. The reason is that, as the critical value is approached, the ratio of the tangential and normal components of the reaction increases without bound. Under real conditions, this value is bounded by the friction coefficient. Experimental testing has shown* [18–20] *that the top will begin to slide before takeoff.*

Note 2. *It is easy to show that similar dynamics arise not only in the limiting case of a point mass on a weightless hoop, but also for values of eccentricity a somewhat smaller than the radius of the sphere* ρ.

3. VARIOUS CASES OF NONHOLONOMIC CONSTRAINTS

In this section, we consider some special cases where nonholonomic constraints are imposed on the top: prohibition of sliding in a certain direction fixed in the body, absolute roughness, and "rubber" ball.

3.1. 3D Analog of Chaplygin's Sleigh [25]

A simple plane nonholonomic system was presented by Chaplygin in 1911 in *Matematicheskii Sbornik*. The translation was made almost a century later by the editors of the journal "Regular and Chaotic Dynamics" [33]. Chaplygin's "sleigh" is a heavy rigid body supported by three points on a horizontal plane and capable of sliding on the support in only one direction, fixed in the body. It is shown that in the general case the system can be integrated by the Hamilton – Jacobi method. We will assume that a similar restriction is imposed on the motion of the top: the velocity of the contact point during sliding is always directed along the projection of the symmetry axis onto the support.

To derive the constraint equation describing this restriction, we express the sliding velocity at the contact point C in terms of generalized coordinates:

$$\vec{v}_C = \vec{v}_G + \vec{\omega} \times \overrightarrow{GC}, \tag{3.1}$$

where $\vec{v}_C$ and $\vec{v}_G$ are the velocities of the contact point and the center of mass, respectively,

$$\begin{gathered} \vec{v}_G = \dot{x}\vec{i}_1 + \dot{y}\vec{i}_2 + \dot{z}\vec{i}_3, \quad \vec{\omega} = p\vec{e}_1 + q\vec{e}_2 + r\vec{e}_3, \\ \overrightarrow{GC} = -(\rho + a\cos\theta)\vec{i}_3 - a\vec{\tau}\sin\theta. \end{gathered} \tag{3.2}$$

Here $\vec{n} = \vec{i}_1\cos\psi + \vec{i}_2\sin\psi$ is the unit vector of the line of nodes and $\vec{\tau} = \vec{n} \times \vec{i}_3 = \vec{i}_1\sin\psi - \vec{i}_2\cos\psi$ is the unit vector of the projection of the top axis onto the support. The relationship between the two coordinate systems is described by the formulas

$$(\vec{e}_1, \vec{e}_2, \vec{e}_3) = (\vec{i}_1, \vec{i}_2, \vec{i}_3)A, \quad A = \|a_{ij}\|, \tag{3.3}$$

$$\begin{gathered} a_{11} = \cos\psi\cos\varphi - \sin\psi\sin\varphi\cos\theta, \quad a_{21} = \sin\psi\cos\varphi + \cos\psi\sin\varphi\cos\theta, \quad a_{31} = \sin\varphi\sin\theta, \\ a_{12} = -\cos\psi\sin\varphi - \sin\psi\cos\varphi\cos\theta, \quad a_{22} = -\sin\psi\sin\varphi + \cos\psi\cos\varphi\cos\theta, \quad a_{32} = \cos\varphi\sin\theta, \\ a_{13} = \sin\psi\sin\theta, \quad a_{23} = -\cos\psi\sin\theta, \quad a_{33} = \cos\theta. \end{gathered}$$

Let us express the angular velocity vector (3.2) in a fixed coordinate system taking into account formulas (2.3) and (3.3):

$$\vec{\omega} = (\dot{\theta}\cos\psi + \dot{\varphi}\sin\theta\sin\psi)\vec{i}_1 + (\dot{\theta}\sin\psi - \dot{\varphi}\sin\theta\cos\psi)\vec{i}_2 + (\dot{\psi} + \dot{\varphi}\cos\theta)\vec{i}_3. \tag{3.4}$$

Substituting this expression into (3.1), we obtain the following equation as a result of elementary but cumbersome calculations:

$$\begin{aligned} \vec{v}_C = \{&\dot{x} - [\rho(\dot{\theta}\sin\psi - \dot{\varphi}\sin\theta\cos\psi) + a\dot{\theta}\cos\theta\sin\psi + a\dot{\psi}\sin\theta\cos\psi]\}\vec{i}_1 \\ &+ \{\dot{y} + [\rho(\dot{\theta}\cos\psi + \dot{\varphi}\sin\theta\sin\psi) + a\dot{\theta}\cos\theta\cos\psi - a\dot{\psi}\sin\theta\sin\psi]\}\vec{i}_2 + (\dot{z} + a\dot{\theta}\sin\theta)\vec{i}_3 \end{aligned} \tag{3.5}$$

(if the unilateral constraint (2.4) is strong, then $\dot{z} + a\dot{\theta}\sin\theta = 0$).

The equation of nonholonomic constraint in the case $(\vec{v}_C, \vec{n}) = 0$ takes the form

$$\dot{x}\cos\psi + \dot{y}\sin\psi - a\dot{\psi}\sin\theta + \rho\dot{\varphi}\sin\theta = 0. \tag{3.6}$$

Thus, in formula (2.2) we obtain

$$l = 1, \quad \gamma_{11} = \cos\psi, \quad \gamma_{12} = \sin\psi, \quad \gamma_{13} = \gamma_{14} = 0, \quad \gamma_{15} = a\sin\theta, \quad \gamma_{16} = -\rho\sin\theta.$$

Therefore, in Eqs. (2.1), the reactions of the nonholonomic constraint (3.6) have the form

$$R_1 = \lambda\cos\psi, \quad R_2 = \lambda\sin\psi, \quad R_3 = R_4 = 0, \quad R_5 = a\lambda\sin\theta, \quad R_6 = -\rho\lambda\sin\theta. \tag{3.7}$$

Let us now turn to LCP (2.5). Formulas (2.9) remain unchanged when (3.7) is taken into account. We come to the conclusion: in the case under consideration, the dynamics have no singularities: $b' = b$ and formula (2.10) is valid. Note that nonzero reactions of the constraint in expressions (3.7) affect the solutions of system (2.1).

3.2. Prohibition of Sliding Along $\vec{\tau}$

Recently, there has been some research [26, 27] on the dynamics of a spherical top on a plane with a prohibition of sliding in the direction $\vec{\tau}$ of the projection of the top's axis onto the support. This means imposing the nonholonomic constraint

$$(\vec{v}_C, \vec{\tau}) = 0. \tag{3.8}$$

Substituting the expression (3.5) into this relation, we obtain

$$\dot{x}\sin\psi - \dot{y}\cos\psi - (\rho + a\cos\theta)\dot{\theta} = 0. \tag{3.9}$$

Hence,

$$R_1 = \lambda\sin\psi, \quad R_2 = -\lambda\cos\psi, \quad R_3 = R_5 = R_6 = 0, \quad R_4 = -\lambda(\rho + a\cos\theta). \tag{3.10}$$

In comparison with the previous formulas (3.7), there is a fundamental difference: now $R_4 \neq 0$, which can lead to a change of the signs of the quantities M, b, b' in GLCP (2.7), i. e., to the emergence of paradoxical situations. Let us study this possibility based on the analysis of Eqs. (2.1).

Note that

$$p^2 + q^2 = \dot{\theta}^2 + \dot{\psi}^2\sin^2\theta, \quad r = \dot{\psi}\cos\theta + \dot{\varphi},$$

from which it follows that the variables ψ and φ are cyclic, and they correspond to the first integrals

$$\frac{\partial T}{\partial\dot{\psi}} = J_1\dot{\psi}\sin^2\theta + J_3 r\cos\theta = k = \text{const}, \quad \frac{\partial T}{\partial\dot{\varphi}} = \dot{\psi}\cos\theta + \dot{\varphi} = r = \text{const}. \tag{3.11}$$

System (2.1) takes the form (assuming $m = 1$)

$$\begin{gathered}\ddot{x} = \lambda\sin\psi, \quad \ddot{y} = -\lambda\cos\psi, \quad \ddot{z} = N - g, \\ \frac{d\left(\left(a^2\sin^2\theta + J_1\right)\dot{\theta}\right)}{dt} - \frac{\partial T}{\partial\theta} = aN\sin\theta - \lambda(\rho + a\cos\theta), \\ k = \text{const}, \quad r = \text{const}.\end{gathered} \tag{3.12}$$

These equations should be considered taking into account the imposed constraints:

$$\dot{x}\sin\psi - \dot{y}\cos\psi - (\rho + a\cos\theta)\dot{\theta} = 0, \quad z = \rho + a\cos\theta.$$

We exclude N from the third of Eqs. (3.12) and substitute the resulting expression into the fourth equation. The latter takes the form

$$\begin{gathered}\left(a^2\sin^2\theta + J_1\right)\ddot{\theta} + 2a^2\dot{\theta}^2\sin\theta\cos\theta - \frac{\partial T}{\partial\theta} = a\left(-a\dot{\theta}^2\cos\theta + g\right)\sin\theta - \lambda(\rho + a\cos\theta), \\ \frac{\partial T}{\partial\theta} = a^2\dot{\theta}^2\sin\theta\cos\theta + J_1\dot{\psi}^2\sin\theta\cos\theta - J_3 r\dot{\psi}\sin\theta.\end{gathered} \tag{3.13}$$

In Eq. (3.13) $\dot{\psi}$ can be expressed from the first integral (3.11), then it will include only the variable θ as well as λ. Eliminating λ from the first two of Eqs. (3.12), we obtain

$$\ddot{x}\cos\psi + \ddot{y}\sin\psi = 0.$$

Taking into account relation (3.9) and the integral (3.11), it follows that

$$\frac{d(\dot{x}\cos\psi + \dot{y}\sin\psi)}{dt} = -(\rho + a\cos\theta)\dot{\psi}\dot{\theta} = -(\rho + a\cos\theta)\frac{k - J_3 r\cos\theta}{J_1\sin^2\theta}\dot{\theta}. \tag{3.14}$$

Integrating this equality, we obtain

$$\begin{gathered}\dot{x}\cos\psi + \dot{y}\sin\psi = -\int\limits_0^\theta (\rho + a\cos\theta)\frac{k - J_3 r\cos\theta}{J_1\sin^2\theta}\,d\theta = G(\theta), \\ G(\theta) = -ar\frac{J_3}{J_1}\theta - \frac{arJ_3 - \rho k}{J_1}\operatorname{ctg}\theta + \frac{ak - J_3 r\rho}{J_1\sin\theta} + B, \quad B = \dot{x}_0\cos\psi_0 + \dot{y}_0\sin\psi_0\end{gathered} \tag{3.15}$$

where the integration constant is determined by the initial conditions.

Next, from the first two of Eqs. (3.12) we express λ:

$$\begin{aligned}\lambda = \ddot{x}\sin\psi - \ddot{y}\cos\psi &= \frac{d(\dot{x}\sin\psi - \dot{y}\cos\psi)}{dt} - (\dot{x}\cos\psi + \dot{y}\sin\psi)\dot{\psi} \\ &= (\rho + a\cos\theta)\ddot{\theta} - a\dot{\theta}^2\sin\theta - G(\theta)\frac{k - J_3 r\cos\theta}{J_1\sin^2\theta}.\end{aligned} \tag{3.16}$$

Substituting the resulting expression into Eq. (3.13), we obtain an equation in a single variable θ:

$$\begin{gathered}\left(a^2 + J_1 + \rho^2 + 2a\rho\cos\theta\right)\ddot{\theta} - J_1\dot{\psi}^2\sin\theta\cos\theta + J_3 r\dot{\psi}\sin\theta = ag\sin\theta - \rho G(\theta)\frac{k - J_3 r\cos\theta}{J_1\sin^2\theta}, \\ b = -g + a\ddot{\theta}\sin\theta + a\dot{\theta}^2\cos\theta - \frac{\lambda(\rho + a\cos\theta)}{J_1}.\end{gathered} \tag{3.17}$$

Note that a similar reduction was carried out in [22]. The quantity b' is determined by formula (2.10).

Having constructed the solution of Eq. (3.17) numerically, we can then determine λ from (3.16) and, finally, M from (2.9); it turns out that the inequality $M > 0$ always holds. An example is shown in Fig. 2. Here the graphs are constructed for the values

$$\begin{gathered}\rho = 1, \quad a = 0.9, \quad J_1 = 0.9, \quad J_3 = 0.2, \quad x_0 = y_0 = 0, \quad \dot{x}_0 = \dot{y}_0 = 0, \\ \psi_0 = 0, \quad \varphi_0 = 0, \quad \dot{\varphi}_0 = 0, \quad \dot{\psi}_0 = 1, \quad \theta_0 = \frac{\pi}{4}, \quad \dot{\theta}_0 = -0.1, \quad g = 1.5.\end{gathered}$$

The solid lines correspond to the graph $\theta(t)$, the dashed and dash-dotted lines denote the graphs $b(t)$ and $b'(t)$ in formulas (2.9), respectively. For the values $t \in (0,\, t_1)$, $t_1 \approx 0.62$, the top moves on the support, and at the point t_1 the finite time paradox occurs (the movement is interrupted).

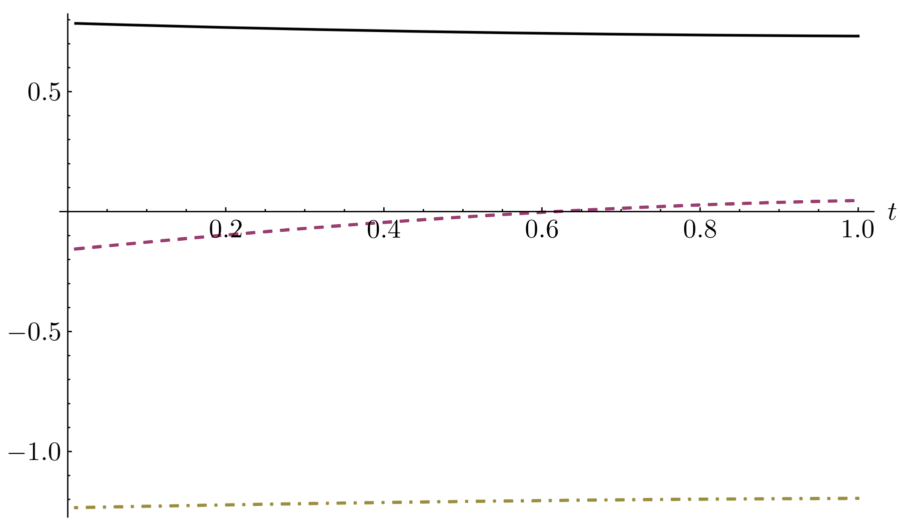

Fig. 2. An example of the finite time paradox: solid line $\theta(t)$, dashed line $b(t)$, dash-dotted line $b'(t)$.

3.3. Complete Absence of Slipping

In the classical formulation, the plane is absolutely rough, i. e., slipping in any direction is impossible [34]. This constraint can be represented as the simultaneous imposition of two nonholonomic constraints (3.6) and (3.9). In this case, system (3.12) is transformed to the form

$$\begin{gathered}\ddot{x} = \lambda_1 \sin\psi + \lambda_2 \cos\psi, \quad \ddot{y} = -\lambda_1 \cos\psi + \lambda_2 \sin\psi, \quad \ddot{z} = N - g, \\ \frac{d\left(\left(a^2 \sin^2\theta + J_1\right)\dot{\theta}\right)}{dt} - \frac{\partial T}{\partial \theta} = aN\sin\theta - \lambda_1(\rho + a\cos\theta), \\ J_1\dot{\psi}\sin^2\theta + J_3(\dot{\psi}\cos\theta + \dot{\varphi})\cos\theta = \lambda_2 a\sin\theta, \\ \dot{\psi}\cos\theta + \dot{\varphi} = \lambda_2\rho\cos\theta.\end{gathered} \tag{3.18}$$

The study is complicated by the presence of two Lagrange multipliers in the system, and so we do not carry it out in this paper. We only note that the existence of the nonuniqueness paradox in this problem was shown in [23].

3.4. The "Rubber" Ball

For such a contact model, along with the prohibition of sliding, there is also no spinning, i. e.,

$$(\omega,\, i_3) = \dot{\psi} + \dot{\varphi}\cos\theta \equiv 0.$$

This leads to the appearance of a third factor in the equations of motion:

$$\begin{gathered}\ddot{x} = \lambda_1 \sin\psi + \lambda_2 \cos\psi, \quad \ddot{y} = -\lambda_1 \cos\psi + \lambda_2 \sin\psi, \quad \ddot{z} = N - g, \\ \frac{d\left(\left(a^2 \sin^2\theta + J_1\right)\dot{\theta}\right)}{dt} - \frac{\partial T}{\partial \theta} = aN\sin\theta - \lambda_1(\rho + a\cos\theta), \\ J_1\dot{\psi}\sin^2\theta + J_3(\dot{\psi}\cos\theta + \dot{\varphi})\cos\theta = \lambda_2 a\sin\theta + \lambda_3, \\ \dot{\psi}\cos\theta + \dot{\varphi} = \lambda_2\rho\cos\theta + \lambda_3\cos\theta.\end{gathered} \tag{3.19}$$

The presence of singularities in the system (3.19) has not yet been studied. It can only be noted that the motion where the finite time paradox arises, as described in Section 2.3, is possible here.

4. DISCUSSION

To date, a large number of application packages have been developed for studying multibody dynamics, and detailed monographs have been written on this issue (e. g. [7, 32]). The most adequate models include the LCP solution for determining the instants of takeoff. However, the GLCP model proposed by the author has not yet become widespread. This paper shows the advantages of such an approach for systems with nonholonomic constraints, but it is equally applicable to systems with rolling and Coulomb friction. For example, an eccentric wheel can take off the support even with a nonzero normal reaction. Ignoring this circumstance in practice can lead to emergency situations. The reason for the finite time paradox is the use of a nonholonomic model in a system with a unilateral constraint, which essentially means that the friction coefficient is unlimited. The singularity can be eliminated if we take into account the finite value of this coefficient. At the same time, the paradox of ambiguity also arises in systems with static friction. The question of its resolution remains open.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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